

Assembling the row from $\mathfrak{A}$ and $\mathfrak{C}$

Does the quoted result follow from the definitions?

$K^+ \equiv p^+ + k^+$ — the first momentum argument of $\mathfrak{C}$

0. Verdict

Yes. All three steps are right — the assembly, the w' integration, and the $+$ prescription. Every factor was checked symbolically:

- numeric prefactor $\frac{1}{4\pi^4}$ — exact;
- overall $-i$ — exact;
- $\frac{1}{k^{+2}}$ — exact, and it arises in a way worth noticing (§2);
- the index bracket $[\delta_{k'm}\delta_{ij} - \frac{p^++k^+}{p^+}\delta_{jm}\delta_{ik'} - \frac{p^++k^+}{k^+}\delta_{im}\delta_{jk'}]$ — exact, all 16 index assignments;
- all four Weizsäcker–Williams kernels and both colour brackets — exact.

1. The coefficient functions

From $-\mathfrak{A}_i^a(k^+, \mathbf{z}) = -\frac{1}{\sqrt{2\pi}} \frac{ig}{\sqrt{k^+}} \int_x \frac{(\mathbf{x}-\mathbf{z})^i}{(\mathbf{x}-\mathbf{z})^2} \rho^a(\mathbf{x})$,

$$\mathfrak{A}(K^+) = \frac{ig}{\sqrt{2}\pi\sqrt{K^+}}, \quad \mathfrak{A}^\dagger(K^+) = \frac{-ig}{\sqrt{2}\pi\sqrt{K^+}},$$

and from the definition of $\mathfrak{C}$, evaluated at $\bar{\mathfrak{C}}_{1jik'}^{\dagger d'ac'}(p^++k^+, y'-w'; p^+, w'-z)$,

$$\mathfrak{C} \longrightarrow -\frac{g f^{c'd'a}}{2\pi\sqrt{2K^+}} \frac{\sqrt{p^+(K^+-p^+)}}{K^+} = -\frac{g f^{c'd'a}}{2\pi\sqrt{2K^+}} \frac{\sqrt{p^+k^+}}{p^++k^+}.$$

There are three $\mathfrak{A}$'s and one $\mathfrak{C}$, so g^4 and, for the phases, $(-i) \times (-1) \times (i) \times (i) = -i$. Multiplying the four numeric factors and the $\frac{2}{(2\pi)^3}$ out front:

$$\frac{2}{(2\pi)^3} \cdot \frac{1}{(\sqrt{2}\pi)^3} \cdot \frac{1}{2\sqrt{2}\pi} = \frac{2}{(2\pi)^3} \cdot \frac{1}{8\pi^4} = \frac{1}{(2\pi)^3} \cdot \frac{1}{4\pi^4} \quad \checkmark$$

The k^+ powers

The $\sqrt{p^+k^+}$ carried by $\mathfrak{C}$ cancels the $\frac{1}{\sqrt{p^+}}$ and $\frac{1}{\sqrt{k^+}}$ of the two $\mathfrak{A}$'s exactly, leaving

$$\frac{1}{\sqrt{K^+}} \cdot \frac{1}{\sqrt{K^+}} \cdot \frac{1}{K^+} \cdot \sqrt{p^+k^+} \cdot \frac{1}{\sqrt{k^+}} \cdot \frac{1}{\sqrt{p^+}} = \frac{1}{(p^++k^+)^2}.$$

2. Where the $\frac{1}{k^{+2}}$ comes from — the delta Jacobian

This is the one step that is easy to lose. $\mathfrak{C}$'s delta is $\delta^{(2)}(z_{\text{arg}} + \frac{p^+}{K^+}x_{\text{arg}})$ with $z_{\text{arg}} = y' - w'$ and $x_{\text{arg}} = w' - z$, i.e.

$$\delta^{(2)}\left(y' - w' + \frac{p^+}{K^+}(w' - z)\right) = \delta^{(2)}\left(-\frac{k^+}{K^+}\left[w' - \frac{(p^++k^+)y'-p^+z}{k^+}\right]\right) = \left(\frac{K^+}{k^+}\right)^2 \delta^{(2)}\left[w' - \frac{(p^++k^+)y'-p^+z}{k^+}\right].$$

The root is the quoted one, but rewriting it in that form costs $\left(\frac{K^+}{k^+}\right)^2$ in two dimensions — and that is exactly what turns $\frac{1}{(p^++k^+)^2}$ into $\frac{1}{k^{+2}}$:

$$\frac{1}{(p^++k^+)^2} \times \frac{(p^++k^+)^2}{k^{+2}} = \frac{1}{k^{+2}} \quad \checkmark$$

Had the Jacobian been dropped, the quoted prefactor would have read $\frac{1}{(p^++k^+)^2}$ instead. It reads $\frac{1}{k^{+2}}$, so it was included correctly.

Assembling everything symbolically,

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product of the four coefficient functions x 2/(2pi)^3 : -i g^4 / (32 pi^7 (k+ + p+)^2)
times the Jacobian (K/k)^2 : -i g^4 / (32 pi^7 k+^2)
quoted -(1/(2pi)^3)(i g^4 f/(4 pi^4))(1/k+^2) : -i g^4 / (32 pi^7 k+^2)
difference : 0

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3. The index bracket

The definition is $\mathfrak{C}_{1jki}$ with $[\delta_{im}\delta_{jk} - \frac{K^+}{p^+}\delta_{jm}\delta_{ik} - \frac{K^+}{K^+-p^+}\delta_{km}\delta_{ij}]$, and the object needed is $\mathfrak{C}_{1jik'}$ — i.e. the relabelling $j \rightarrow j, k \rightarrow i, i \rightarrow k'$. Under it

$$\delta_{im}\delta_{jk} \rightarrow \delta_{k'm}\delta_{ij}, \quad \delta_{jm}\delta_{ik} \rightarrow \delta_{jm}\delta_{ik'}, \quad \delta_{km}\delta_{ij} \rightarrow \delta_{im}\delta_{jk'},$$

and with $K^+ = p^+ + k^+$ so that $K^+ - p^+ = k^+$,

$$\left[\delta_{k'm}\delta_{ij} - \frac{p^+ + k^+}{p^+} \delta_{jm}\delta_{ik'} - \frac{p^+ + k^+}{k^+} \delta_{im}\delta_{jk'} \right] \quad \checkmark$$

Checked over all 16 index assignments: zero mismatches.

4. Kernels and colour

factor	gives	colour
$\mathfrak{C}'s \frac{x^m}{x^2}, x = w' - z$	$\frac{(w' - z)^m}{(w' - z)^2}$	$f^{c'da}$
$\mathfrak{A}_{k'}^{i' e'}(K^+, -y')$	$\frac{(x' - y')^{k'}}{(x' - y')^2} \rho^{e'}(x')$	$U^{e'c'}(y') U^{d'b}(z) / U^{bd'}(z) U^{c'e'}(x')$
$\mathfrak{A}_i^a(k^+, -w)$	$\frac{(y - w)^i}{(y - w)^2}$	$U^{ad}(y) \rho^d(y)$
$\mathfrak{A}_j^c(p^+, -z), \mathfrak{A}_j^b(p^+, -z)$	$\frac{(x - z)^j}{(x - z)^2}$	$U^{bc}(z) U^{ce}(x) \rho^e(x) / \rho^b(x)$

which reproduces the quoted numerator and both colour brackets exactly.

5. The two later steps

The w' integration. Verified previously: the delta sets $w' = y' + \frac{p^+}{k^+}(y' - z)$, so $w' - z = \frac{p^+ + k^+}{k^+}(y' - z)$, the first kernel supplies $\frac{k^+}{p^+ + k^+}$, the phase splits as $e^{-ik \cdot (y' - w)} e^{-i \frac{p^+}{k^+} k \cdot (y' - z)}$, and pulling $(p^+ + k^+)$ out of the bracket leaves $\frac{1}{k^+} \left[\frac{\delta_{k'm}\delta_{ij}}{p^+ + k^+} - \frac{\delta_{im}\delta_{jk'}}{p^+} - \frac{\delta_{jm}\delta_{ik'}}{k^+} \right]$. Correct.

The $+$ prescription. Verified previously: sign, prefactor $\frac{1}{8\pi^5} = \frac{1}{4\pi^4} \cdot \frac{1}{2\pi}$, the $\delta_{jm}\delta_{ik'}$ contraction $(y' - z) \cdot (x - z) (x' - y') \cdot (y - w)$, the dropped phase, and keeping the lower limit at Λ , which makes the rearrangement *exact*. Correct.

Two things not derived from the definitions, because they cannot be — they are inputs: the limits $\int_{\Lambda}^{\vee - k^+} dp^+$ (the rapidity cutoffs you impose), and the convention that $\mathfrak{A}(K^+, -\mathbf{z})$ carries the kernel $\frac{(x - \mathbf{z})^i}{(x - \mathbf{z})^2}$ rather than $\frac{(x + \mathbf{z})^i}{(x + \mathbf{z})^2}$. Both are consistent throughout.

And the standing caveat: the first term of the last equation is the soft-subtracted remainder, not a finite contribution — it still diverges at large $|y' - z|$.

zint/src/assemble_row.py — every check above, written to zint/RESULTS_assemble.txt.